<table>
<tr><th>Question</th><th>Answers</th><th>Extra information</th><th>Mark</th><th>AO / Spec. Ref.</th></tr>
<tr><td>05.1</td><td>
<table>
<tr><th>Factor</th><th>Biotic</th><th>Abiotic</th></tr>
<tr><td>Nitrates in the soil</td><td></td><td>✓</td></tr>
<tr><td>Rabbits eating the plants</td><td>✓</td><td></td></tr>
<tr><td>Shading by a building</td><td></td><td>✓</td></tr>
<tr><td>Soil pH</td><td></td><td>✓</td></tr>
<tr><td>Temperature</td><td></td><td>✓</td></tr>
<tr><td>Trampling by people</td><td>✓</td><td></td></tr>
</table>
</td><td>all 6 correct = 3 marks
4 or 5 correct = 2 marks
2 or 3 correct = 1 mark
0 or 1 correct = 0 marks</td><td>3</td><td>AO1
4.7.1.2
4.7.1.3
4.4.1.2</td></tr>
<tr><td>05.2</td><td>(grid and) coordinates

to achieve randomness</td><td>

ignore throwing quadrat

allow random coordinates for 2 marks

if no other mark awarded allow random walk or description of random walk for 1 mark</td><td>1

1</td><td>AO1
4.7.2.1
RPA9</td></tr>
</table>

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	*(mean per m^2 =)* 24 **or** 6 × 4		1	AO2 4.7.2.1 RPA9
	(calculation of area of lawn =) (½ × 16 × 10) – (6 × 3) **or** 80 – 18		1	
	(area of lawn =) 62 m^2	allow correct calculation using total area (of triangle) – area of rectangle	1	
	(total number of daisies =) 24 × 62	allow correct calculation using an incorrectly calculated area of the lawn and / or mean	1	
	1488	allow answer based on incorrect area	1	
	(answer to 3 sig figs =) 1490	allow student's calculated answer rounded to 3 sig figs	1	
05.4	too few quadrats **or** quadrat too small	allow sample size too small	1	AO3 4.7.2.1 RPA9
	sample may not be representative of the lawn	allow quadrats may not have been placed randomly	1	
Total			**13**	